

$$(2) \quad \nabla_z f(x) \approx \frac{f(x+\delta z) - f(x-\delta z)}{2\delta}$$

$$\nabla f(x) \approx (\nabla_z f(x)) z$$

$$z = [1, -1, 1], \quad \delta = 0.5$$

$$x + \delta z = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix} + 0.5 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1.5 \\ -1.5 \\ 3.5 \end{bmatrix}$$

$$x - \delta z = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix} - 0.5 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.5 \\ -0.5 \\ 2.5 \end{bmatrix}$$

$$f(x + \delta z) = 1.5 - 4.5 - 24.5 = -27.5$$

$$f(x - \delta z) = 0.5 + 3(-0.5) - 2(2.5)^2 = 0.5 - 1.5 - 12.5 = -13.5$$

$$\nabla_z f(x) \approx \frac{-27.5 + 13.5}{2(0.5)} = -14$$

$$\text{Estimate} \quad \nabla f(x) \approx -14 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -14 \\ 14 \\ -14 \end{bmatrix}, \quad \nabla f\left(\begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} -14 \\ 14 \\ -14 \end{bmatrix}$$

$$\text{Exact} \quad \nabla f(x) = \begin{bmatrix} 1 \\ 3 \\ -4x_3 \end{bmatrix}, \quad \nabla f\left(\begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 3 \\ -12 \end{bmatrix}$$

(3) For n queries:

$$I_n = \frac{I_1}{\varphi^{n-1}}, \quad I_1 = [a, b] \leftarrow \text{original} \quad \varphi = 1.62$$

$$I_6 = \frac{50 - 20}{(1.62)^5} \approx 2.69$$

$$\textcircled{4} \quad \eta = \frac{f(x) - f(x')}{f(x) - \hat{f}(x')} \quad \begin{array}{l} f(x) = 10 \\ f(x') = 7.6 \end{array}, \quad H = 2I$$

$$\frac{1}{2} (x' - x)^T H (x' - x) = \|x' - x\|^2 \quad \text{and} \quad \|x' - x\|^2 \leq 5$$

$$\hat{f}(x') = f(x) + g^T (x' - x) + \frac{1}{2} (x' - x)^T H (x' - x) = 10 + g^T (x' - x) + \|x' - x\|^2$$

Minimizing $\hat{f}(x')$, with $x = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$\nabla_x \hat{f}(x') = g + 2x' = 0 \Rightarrow x' = -\frac{1}{2} g = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\|x' - x\| = \sqrt{4+1} = \sqrt{5} < 5$$

$$\text{Then } x' = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\hat{f}(x') = 10 + [-4 \ -2] \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \|x' - x\|^2 = 10 - 10 + 5 = 5$$

$$\eta = \frac{10 - 7.6}{10 - 5} = 0.48$$

Candidate point is good enough as $\eta_1 < \eta < \eta_2$

$$\textcircled{5} \quad \beta^{(k)} = \frac{\begin{bmatrix} -1 & 2 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix}}{\begin{bmatrix} 2 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \end{bmatrix}} = \frac{5}{20} = \frac{1}{4}$$

$$d^{(k)} = - \begin{bmatrix} -1 \\ 2 \end{bmatrix} + \frac{1}{4} d^{(k-1)} = - \begin{bmatrix} -1 \\ 2 \end{bmatrix} + \frac{1}{4} \left(- \begin{bmatrix} 2 \\ 4 \end{bmatrix} \right) = \begin{bmatrix} \frac{1}{2} \\ -3 \end{bmatrix}$$

$$\textcircled{6} \quad f''(x^{(k)}) \approx \frac{f'(x^{(k)}) - f'(x^{(k-1)})}{x^{(k)} - x^{(k-1)}}$$

$$f''(x^{(2)}) \approx \frac{1 - (-3)}{4 - 2} = 2$$

$$x^{(3)} = 4 - \frac{4 - 2}{1 - (-3)} \cdot 1 = 3.5$$

$$\textcircled{7} \quad J^{(n)} = x^{(n+1)} - x^{(1)}$$

$$x^{(1)} = [0, 0]$$

$$J^{(3)} = \begin{bmatrix} 3 \\ 4 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \underline{\begin{bmatrix} 3 \\ 4 \end{bmatrix}}$$

$$x^{(3)} = [3, 4]$$